Requirements for CRM Build (Product Name “The Hestia Suite”)

1. Hardware Requirements (The Host Machine)

Since you are only serving **1–10 people**, your CRM's resource footprint will be incredibly small. You do not need a beefy server.

- **Processor:** 4 Cores (Any modern Intel Core i5/i7 or AMD Ryzen/EPYC).
- **RAM:** 8 GB minimum (**16 GB highly recommended** to smoothly run Docker, Postgres, and the Cloudflare Tunnel daemon simultaneously).
- **Storage:** 50 GB+ NVMe or SSD (HDDs will feel sluggish when processing database queries).
- **OS:** Clean installation of **Ubuntu 24.04 LTS Server** (or Desktop if you prefer a visual GUI).

2. Software & Account Prerequisites

Before prompt-engineering your CRM with Claude, you need to set up your accounts and install the base engine on Ubuntu:

Accounts Needed (All Free Tiers)

- **GitHub Account:** Acts as the bridge. Lovable or Cursor will push code changes to GitHub, and your Ubuntu machine will pull them down.
- **Cloudflare Account:** Required to set up the secure Zero Trust Tunnel. You will also need a custom domain (e.g., yourcompany.com) pointed to Cloudflare's nameservers.
- **Supabase Account (Optional but Smart):** If you want an off-site, cloud-managed database that has a generous free tier. Alternatively, you can host Postgres locally in Docker.

Ubuntu Server Packages to Install

Run this quick command in your Ubuntu terminal to ensure your system has the core engines ready:

Bash

```
None
sudo apt update && sudo apt upgrade -y
sudo apt install -y curl git docker.io docker-compose
```

3. The CRM Architecture Blueprint

This is how the application flows. You will tell the AI developer (Lovable or Cursor) to build a **Vite + React frontend** and a **PostgreSQL backend**.

When the app is complete, you will move the files onto your Ubuntu machine and use the following Docker architecture.

The docker-compose.yml Template

Create a directory on your Ubuntu server (e.g., /home/ubuntu/crm/) and save this file. **This is the exact file you can hand to Claude to populate with your specific database keys:**

YAML

```
None
version: '3.8'

services:
  # 1. The CRM Frontend & API Server (Built from Lovable/Cursor
output)
  crm-app:
    build:
      context: ./frontend
      dockerfile: Dockerfile
    container_name: crm_frontend
    restart: always
    environment:
      -
DATABASE_URL=postgres://crm_user:SecurePassword123@crm-db:5432/crm_
prod
    depends_on:
      - crm-db
    # Note: We DO NOT expose ports to the host. Cloudflare will
talk to it internally.

  # 2. The Local Database
  crm-db:
    image: postgres:16-alpine
    container_name: crm_database
    restart: always
    environment:
      POSTGRES_USER: crm_user
```

```
      POSTGRES_PASSWORD: SecurePassword123
      POSTGRES_DB: crm_prod
    volumes:
      - postgres_data:/var/lib/postgresql/data

  # 3. The Secure Cloudflare Tunnel (No open router ports!)
  cloudflare-tunnel:
    image: cloudflare/cloudflared:latest
    container_name: cloudflare_tunnel
    restart: always
    command: tunnel --no-autoupdate run --token
YOUR_CLOUDFLARE_TUNNEL_TOKEN_HERE

volumes:
  postgres_data:
```

4. Step-by-Step Execution Workflow

To successfully build and deploy this app, follow this workflow with Claude:

Step A: Build the Application UI (In the cloud)

1. Go to **Lovable.dev** or open **Cursor**.
2. Prompt the AI: *"Build a lightweight CRM for a 10-person sales team. It needs a contacts tab, a pipeline view with drag-and-drop Kanban cards, and interaction timelines."*
3. Once the AI builds a working app, use the built-in **GitHub Integration** to export the entire codebase to a private repository on your GitHub account.

Step B: Connect your Ubuntu Server to GitHub

On your Ubuntu machine, clone your newly created repository:

Bash

```
None
git clone https://github.com/yourusername/your-crm-repo.git
/home/ubuntu/crm/frontend
```

Step C: Set up the Cloudflare Tunnel

1. Log into your **Cloudflare Dashboard** and navigate to **Zero Trust** -> **Networks** -> **Tunnels**.
2. Create a new tunnel named `ubuntu-crm`.

3. Cloudflare will display a long token (YOUR_CLOUDFLARE_TUNNEL_TOKEN_HERE). Paste this token directly into your docker-compose.yml file on Ubuntu.
4. In the Cloudflare Tunnel dashboard, route a public hostname (e.g., crm.yourcompany.com) to the internal service: http://crm-app:3000 (or whatever port your app frontend runs on).

Step D: Launch!

Navigate to your folder on your Ubuntu machine and fire up the system:

Bash

```
None
cd /home/ubuntu/crm/
sudo docker-compose up -d --build
```

Within 60 seconds, your entire team will be able to securely type crm.yourcompany.com into their phones, laptops, or home computers and access your locally-hosted, custom-built CRM—completely free of monthly hosting charges.